

Motivation: Study sheaves of dg-algebras (derived rings) on dg varieties + applications.

Specifically the derived category.

$(X, \mathcal{O}_X)$ ringed space.

A sheaf of dg- $\mathcal{O}_X$ -algebras on X is a chain complex A of $\mathcal{O}_X$ -modules with $\mathcal{O}_X$ -bilinear maps

$$\mu: A^n \times A^m \longrightarrow A^{n+m}$$

- (1) μ is associative
- (2) $\exists$ a global section 1 of A^0 that is a 2-sided identity for μ .
- (3) for $U \subseteq X$ open

$$a \in A^n(U), b \in A^m(U)$$

$$d^{n+m}(\mu(ab)) = d^n(a)b + (-1)^n a d^m(b).$$

dg-A-module is a cochain complex $\mathcal{M}^\bullet$ endowed with $\mathcal{O}_X$ -bilinear maps

$$\mu_n : \mathcal{M}^n \times A^m \rightarrow \mathcal{M}^{n+m}$$

- (1) μ_n is associative.
- (2) the identity 1 of A^0 acts as identity on $\mathcal{M}^n \forall n$.
- (3) for $U \subseteq X$ open, $x \in \mathcal{M}^n(U)$
 $a \in A^m(U)$

$$d^{n+m}(xa) = d^n(x)a + (-1)^n x d^m(a).$$

$\text{Mod}(A)$, $D(A)$, $D^b(A)$ defined naturally.

Ex: (1) Sheaves of $\mathcal{O}_X$ -algebras.

(2) $(X, \mathcal{O}_X)$

$$\phi: \mathcal{E} \longrightarrow \mathcal{O}_X$$

$K(\phi)$: (a) as an algebra it is $\wedge \mathcal{E}$.

(b) $d: K_*(\phi) \rightarrow K_*[\phi]$ unique derivation s.t.

$$d_1 = \phi.$$

(3) Dolbeault complex on a complex manifold.

(not $\mathcal{O}_X$ -linear, $\mathbb{C}$ -linear)

(4) $\text{End}(F)^{\rightarrow}$ complex of sheaves.

(5) $\pi: X \xrightarrow{\text{proper}} \text{Spec}(R)$
(Noeth)

$$R\pi_* \mathcal{O}_X.$$

Approach: "noncommutative"

"downside" = outside to projective
variety.

"upside" = addresses n cases.

$$A = \bigoplus_{n \geq 0} A_n \quad \text{graded algebra}$$
$$A_0 = k.$$

[Artin - Schelter, Tate - Van den Bergh]

n analogues of $\mathbb{P}^n$
($\mathbb{P}^2??$)

AS- Regular algebra: (1) A has finite
global dimension.

(2) $H_A(n) = O(f(n))$ f is a polynomial.

(3) $\text{Ext}_A^*(k, A) \cong k$.

$\mathbb{R} \times \mathbb{Q}$ (1) A commutative + AS-regular

$\Downarrow$
weighted poly. ring.

$$(2) \quad |p| - k_q[x, y] = \frac{k\langle x, y \rangle}{(xy - qxy)} \\ q \in \mathbb{R}^\times.$$

[Artin - Zhang]

$$qgr A := \frac{\text{mod-} A}{\text{tors } A}$$

$\searrow$ behaves like $\text{Coh}(X)$.

A comm. gen in degree 1,

$$\text{Coh}(\text{Proj } A) \cong qgr A$$

$$F \longmapsto \bigoplus_{d=-\infty}^{\infty} H^0(X, F \otimes \mathcal{O}_X(d))$$

Satisfactory Cohomology theory if

A satisfies $\mathcal{X}$ if $\forall M \in \text{Mod-}A$
and $\forall n$

$$\underline{\text{Ext}}_A^m(k, M) = 0 \quad \text{for } m \gg 0$$

See vanishing for $\text{qgr}(A)$.

X alg variety

$D^b(X)$ = bounded complexes of
coherent sheaves on X

ω / quasi-invertible

A algebra

$D^b(A)$ = bounded complexes of
f.g. right A -modules

ω / quasi-invertible.

$$\operatorname{Hom}_{D^b(X)}(\mathcal{O}, \mathcal{F}) = H^0(X, \mathcal{F})$$

$$\underline{E}_X: X = \mathbb{P}_{\mathbb{C}}^n$$

$$H^*(\mathbb{P}^n; \mathbb{Z}) \cong \mathbb{Z}^{n+1}$$

$$K_0(\mathbb{P}^n) \cong \mathbb{Z}^{n+1}$$

Both can be recovered from $D^b(\mathbb{P}^n)$.

Semi-orthogonal decomposition

$$\mathcal{D}^b(X) = \langle T', T'' \rangle \text{ if}$$

- (a) T', T'' generate $\mathcal{D}^b(X)$.
- (b) $\text{Hom}(T'', T') = 0$
- (c) $T'' \hookrightarrow \mathcal{D}^b(X)$ has a right adjoint
 $T' \hookrightarrow \mathcal{D}^b(X)$ has a left adjoint.

Ex: X smooth

$$K_0(X) = K_0(\mathcal{D}^b(X)) = K_0(T') \oplus K_0(T'').$$

Full exceptional collection

Objects $T_0, \dots, T_n \in \mathcal{D}^b(X)$

$$(a) \text{Hom}_T(T_i, T_j[n]) = \begin{cases} \mathbb{A} & \text{if } n=0 \\ 0 & \text{otherwise} \end{cases}$$

(b) whenever $j > i$

$$\text{Hom}(T_j, T_i[n]) = 0$$

$$\forall n \in \mathbb{Z}.$$

(c) $T_0, \dots, T_n$ generate $D^b(X)$.

Ex: X smooth

$$K_0(X) \cong \mathbb{Z}^{n+1}.$$

Thm [Beilinson]

$$D^b(\mathbb{P}^n) = \langle \overbrace{O, \dots, O(n)}^{\text{exceptional}} \rangle$$

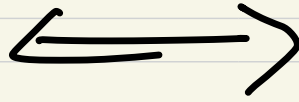
$$\Rightarrow K_0(\mathbb{P}^n) \cong \mathbb{Z}^{n+1}.$$

Thm [Bouvier]

Beilinsons $\longrightarrow$ projective complete intersections.
(CI)

X projective CI

Coherent sheaves
on
 X



Singularity
of
 $\text{cone}(X)$

Singularity category

A algebra

$$D^{\text{sing}}(A) = D^b(A) / \text{Perf}(A)$$

Thm (Culver, special case)

$$S = \mathbb{C}[x_0, \dots, x_n] \quad \deg(x_i) = 1$$

$f_1, \dots, f_c \in S$ regular sequence

$$X := V(f_1, \dots, f_c) \subseteq \mathbb{P}^n.$$

$$R = S/(f_1, \dots, f_c)$$

$$\omega_X \cong G_X(-a)$$

$\swarrow$ $\times$ smooth
 $\searrow$ $\text{dim } X$
 $\wedge$ Ω_X

Then

a) if $a > 0$ (Fano)

$$D^b(X) = \langle \overbrace{G(-a+1), \dots, 0}^{\text{exceptional}}, D^{\text{reg}}(R) \rangle$$

b) if $a < 0$ (general type)

$$D^{\text{reg}}(R) = \langle \overbrace{G, \dots, G(a+1)}^{\text{exceptional}}, D^b(X) \rangle$$

c) if $a = 0$ (Calabi-Yau)

$$D^h(X) \cong D^{\text{sing}}(R)$$

Cor $R = \mathbb{C}$ Beilinson's Theorem

Ex: $X = \{x^3 + y^3 + z^3 = 0\} \subseteq \mathbb{P}^2$

$$D^h(X) \cong \text{Matrix factorizations of } x^3 + y^3 + z^3 \text{ over } \mathbb{C}[x, y, z]$$

homotopy.

Associated to an embedding

$X \hookrightarrow \mathbb{P}^n$, there is a map

$$\phi: \bigoplus_{i=1}^s \mathcal{O}_{\mathbb{P}^n}(d_i) \rightarrow \mathcal{O}_{\mathbb{P}^n}$$

Set $\mathcal{K} := \mathcal{K}_\bullet(\phi)$.

$D^b(\mathcal{K}) =$ sheaves of dg $\mathcal{K}$ -modules on $\mathbb{P}^n$.

$f_i \in \Gamma(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(d_i))$ defining ϕ .

S hom. coordinate ring of $\mathbb{P}^n$.

$$\mathcal{K} := \left(\bigwedge \bigoplus_{i=1}^s S(d_i), \underbrace{d} \right)$$

unique derivation

such that d_i looks like

$$\bigoplus_{i=1}^s S(d_i) \xrightarrow{(f_1 - f_2)} S$$

$$D^{\text{sig}}(K) = D^b(K) / \text{Perf}(K)$$

Thm [Brown-Siebert, special case]

Consider an embedding $X \hookrightarrow \mathbb{P}^n$

g_K $f_1, \dots, f_c$ and K as above.

$$\text{Set } a := n+1 - \sum_{i=1}^c \deg(f_i).$$

a) if $a > 0$ exceptional.

$$D^b(g_K) = \langle g_K(-a+1), \dots, g_K, D^{\text{sig}}(K) \rangle$$

b) if $a < 0$

$$D^{\text{sig}}(K) = \langle \underbrace{\mathbb{C}, \dots, \mathbb{C}(a+1)}_{\text{enough}}, D^b(g_K) \rangle$$

c) if $a = 0$

$$D^{\text{sig}}(K) \simeq D^b(g_K).$$

application to re Hodge theory

- A proper connected Cw dg-algebra.

New cases of Lattice Conjecture:

$$D^b(A)_{dg}, D^b(A)_{dg}^{enrg.}$$

X compact top space

$$Ch: K_{\mathbb{Q}}(X) \otimes \mathbb{Q} \xrightarrow{\cong} H^{even}(X; \mathbb{Q})$$

$\mathcal{C}$ smooth proper dg-category

$$Ch^{top}: K_{\ast}^{top}(\mathcal{C}) \otimes \mathbb{C} \longrightarrow HP_{\ast}(\mathbb{C})$$

Lattice Conj: Ch^{top} is an iso.

A a homogeneous dg- k -algebra

$A = \bigoplus_{i,j \in \mathbb{Z}^2} A_{i,j}^{\vee}$ equipped with a degree $(0,1)$ differential d_A .

and satisfies the Leibniz rule.

$A_0 = A_0^0 = k$. $A_{i,j}^{\vee} = 0$ if $i < 0$
or $j > 0$

$D(A)$,

$D_{\text{coh}}(A) = D^b(A) / D_{\text{tor}}(A)$

behaves like $D^b(A)$, A a sheaf of dg-algebras on a space X .

field
lin
homologie.

A^0 commutative

$D_{\text{coh}}(A) =$ derived cat of sheaves
of dg-algebras on
 $\text{Proj}(A^0)$.

$$A \longleftrightarrow H^0(A)$$

Duality theory $\longleftrightarrow$ duality theory
exists for $H^0(A)$.

$$M \in D^b(A)$$

$$\varinjlim_A \text{Ext}_A^n(k, M) = 0 \quad m \gg 0$$

~~for~~
 $\forall n$.

$$\widetilde{(-)} : D^h(A) \rightarrow D^b(A) / D_{\text{acy}}(A)$$

$$H^j(\text{Coh}(A), \widetilde{M}) \cong \text{Ext}^{-j-1}(\widetilde{\mathcal{O}}, \widetilde{\mathcal{R}})^*$$

$\downarrow$
 balanced
 dualizing
 dg-module.